# Volume 3: The Physics — Index

2026-05-19

## Volume 3: The Physics — Index

Target reader: Engaged physicist working with the framework; wants full BST physics

Voice/style: Technical, dynamics-level

### Chapters

| Chapter | Topic | Contents |
|---|---|---|
| `Ch01_Gravity_Vacuum.md` | Gravity and Vacuum | Sections 10-12: Gravity as Statistical Thermodynamics (Newton's G), Chiral Condensate, Vacuum Energy |
| `Ch02_Quantum_Interface.md` | Quantum-Classical Interface (Hamiltonian + Bergman Dirac) | Section 13: includes Hamiltonian $H = (\text{winding})^2$, $\hbar = 2mD$, Born Rule from Gleason, 13.9 Bergman Dirac Operator (Spring 2026 LAG-1 Sessions 1-10 full operator-level work, eigentones $\lambda$=10 + $\lambda$=75/4) |
| `Ch03_Forces_Cosmology.md` | Forces and Cosmology | Sections 14-21: Three Geometric Layers, Cosmological Implications, Matter Clumping, Information Theory, 2D-to-3D Interface, Dark Matter as Channel Noise, Weak Force as Variation Operator, Thermodynamic Foundation |
| `Ch04_Cosmic_Cascade.md` | Cosmic Architecture | Sections 22-25: Arrow of Time, Wavefront, Growing Manifold, Cascade of Forced Choices |
| `Ch05_Discussion.md` | Discussion — Lagrangian Status + Partition Function + Central Claim | Section 26: Six-term Lagrangian S_BST status, Partition Function as Master Calculation, QM and GR as scale limits of one substrate, Arrow of Complexity |

### Supporting computational subdirectory

- `rotation_curves/` — SPARC galaxy rotation curve data + `sparc_bst.py` analysis script + `DarkMatter-Calculation.md` documenting BST's channel-noise framework for galaxy rotation curves (no dark matter particles required). Directly supports Chapter 3 (Dark Matter as Channel Noise, originally Section 19 of monolithic). Moved into Volume 3 on 2026-05-19 from repository root.

## Build instructions

Build the volume PDF with pandoc:

```
cd Working_Paper/Vol3_Physics
pandoc Ch*.md -o Volume.pdf --pdf-engine=xelatex -H ../notes/bst_pdf_header.tex
```

## Navigation

- Up: Master Library Index
- All volumes: Vol 1 Journey, Vol 2 Framework, Vol 3 Physics, Vol 4 Mathematics, Vol 5 Predictions, Vol 6 Frontier

— Keeper, 2026-05-19